

MATH3018 Assignment 4, due on Mar 5

1. Prove the following [3pts]: (a). If $\mathbf{M} \sim W_p(\boldsymbol{\Sigma}, n)$ and $\mathbf{A}$ is a fixed $q \times p$ matrix, then

$$\mathbf{A}\mathbf{M}\mathbf{A}^\top \sim W_q(\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top, n).$$

- (b). If $\mathbf{M} \sim W_p(\boldsymbol{\Sigma}, n)$ and $\mathbf{a}$ is a fixed $p \times 1$ vector then

$$\mathbf{a}^\top \mathbf{M} \mathbf{a} \sim (\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}) \chi_n^2.$$

Solution

- (a) From the definition, let $\mathbf{M} = \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top$, where $\mathbf{x}_i \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$. Then

$$\begin{aligned} \mathbf{A}\mathbf{M}\mathbf{A}^\top &= \mathbf{A} \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \right) \mathbf{A}^\top \\ &= \sum_{i=1}^n (\mathbf{A}\mathbf{x}_i) (\mathbf{A}\mathbf{x}_i)^\top = \sum_{i=1}^n \mathbf{y}_i \mathbf{y}_i^\top \end{aligned}$$

where $\mathbf{y}_i = \mathbf{A}\mathbf{x}_i \sim N_q(\mathbf{0}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top)$. Now we apply the definition of the Wishart distribution to $\mathbf{y}_1, \dots, \mathbf{y}_n$ and, hence, $\sum_{i=1}^n \mathbf{y}_i \mathbf{y}_i^\top \sim W_q(\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top, n)$.

- (b) Applying the conclusion in (a) with $\mathbf{A} = \mathbf{a}^\top$, we see $\mathbf{a}^\top \mathbf{M} \mathbf{a} \sim W_1(\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}, n)$. If we let $z_i \sim N(0, 1)$, and $\sigma = (\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})^{\frac{1}{2}}$, then $\sigma z_i \sim N(0, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$. Thus

$$\begin{aligned} W_1(\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}, n) &\sim \sum_{i=1}^n \sigma^2 z_i^2 \quad \text{by the definition of the Wishart distribution} \\ &= \sigma^2 \sum_{i=1}^n z_i^2 \\ &\sim (\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}) \chi_n^2 \quad \text{by the definition of } \chi^2 \end{aligned}$$

2. Let $\mathbf{x}_1, \dots, \mathbf{x}_{10}$ be a random sample of size $n = 10$ from an $N_4(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ population. Find the following (directly write down the conclusions) [5pts]:

- The distribution of $(\mathbf{x}_1 - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_1 - \boldsymbol{\mu})$.
- The distribution of $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$.
- The distribution of $\sqrt{n}(\bar{\mathbf{x}} - \boldsymbol{\mu})$.
- The distribution of $(n-1)\mathbf{S}$, where $\mathbf{S}$ is the sample covariance matrix.
- The distribution of $B(9\mathbf{S})B^\top$, where

$$B = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Solution (a) χ_4^2 . (b) $N_4(\mu, \frac{1}{10}\Sigma)$. (c) $N_4(0, \Sigma)$. (d) $W_4(9, \Sigma)$. (e) $W_2(9, B\Sigma B^T)$, where

$$B\Sigma B^T = \begin{pmatrix} \sigma_{11} & \sigma_{13} \\ \sigma_{31} & \sigma_{33} \end{pmatrix}$$

3. (a) Evaluate T^2 , for testing $H_0 : \mu^T = (7, 11)$, using the data

$$\mathbf{X} = \begin{pmatrix} 2 & 12 \\ 8 & 9 \\ 6 & 9 \\ 8 & 10 \end{pmatrix}$$

[1pt]

- (b) What is the distribution of Hotelling's T^2 under H_0 ? [1pt]

- (c) Test H_0 at the $\alpha = 0.05$ level. [1pt]

Solution. (a) We have $n = 4, p = 2, \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \begin{pmatrix} 6 \\ 10 \end{pmatrix}$, and

$$S = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^T = \begin{pmatrix} 8 & -\frac{10}{3} \\ -\frac{10}{3} & 2^{\frac{2}{3}} \end{pmatrix} \Rightarrow S^{-1} = \begin{pmatrix} \frac{9}{22} & \frac{15}{22} \\ \frac{15}{22} & \frac{11}{11} \end{pmatrix}$$

$$T^2 = n(\bar{x} - \mu)^T S^{-1}(\bar{x} - \mu) = 13.6364$$

- (b) Under H_0, T^2 is distributed as $\frac{(n-1)p}{n-p} F_{p, n-p}$. That is,

$$T^2 \sim 3F_{2,2}$$

- (c) Using R we calculate $F_{2,2}(0.05) = 19$. Since $T^2 = 13.6364 < 3F_{2,2}(0.05) = 57$, we do not reject H_0 at significant level $\alpha = 0.05$.

4. Solve Problem 5.7 of the reference book "Applied Multivariate Statistical Analysis" by Richard A. Johnson and Dean W. Wichern (six edition). (The data is given below, and also included in the Excel file "Sweat-Data.csv", use R to do necessary calculations, but providing R code is not required) [4pts]

Solution

By calculations,

$$\bar{x} = \begin{bmatrix} 4.64 \\ 45.4 \\ 9.965 \end{bmatrix}, S = \begin{bmatrix} 2.879 & 10.01 & -1.809 \\ 10.01 & 199.7884 & -5.64 \\ -1.809 & -5.64 & 3.628 \end{bmatrix}.$$

For μ_i , the T^2 confidence intervals are

$$\left[\bar{x}_i - c\sqrt{\frac{s_{ii}}{n}}, \bar{x}_i + c\sqrt{\frac{s_{ii}}{n}} \right]$$

where $c^2 = \frac{(n-1)p}{n-p} F_{p,n-p}(0.05) = \frac{19 \times 3}{17} F_{3,17}(0.05) = \frac{19 \times 3}{17} \times 3.2 = 10.73$. So

$$\begin{aligned}\mu_1 &: [4.64 - \sqrt{10.73} \cdot \sqrt{2.879/20}, 4.64 + \sqrt{10.73} \cdot \sqrt{2.879/20}] = [3.397, 5.883] \\ \mu_2 &: [45.4 - \sqrt{10.73} \cdot \sqrt{199.7884/20}, 45.4 + \sqrt{10.73} \cdot \sqrt{199.7884/20}] = [35.047, 55.753] \\ \mu_3 &: [9.965 - \sqrt{10.73} \cdot \sqrt{3.628/20}, 9.965 + \sqrt{10.73} \cdot \sqrt{3.628/20}] = [8.570, 11.360]\end{aligned}$$

The Bonferroni intervals are

$$\left[\bar{x}_1 - t_{n-1} \left(\frac{\alpha}{2p} \right) \sqrt{\frac{s_{ii}}{n}}, \bar{x}_1 + t_{n-1} \left(\frac{\alpha}{2p} \right) \sqrt{\frac{s_{ii}}{n}} \right]$$

where we use $t_{19}(0.05/6) = 2.625$, so

$$\begin{aligned}\mu_1 &: [3.644, 5.635] \\ \mu_2 &: [37.103, 53.697] \\ \mu_3 &: [8.847, 11.083]\end{aligned}$$

We can see that the Bonferroni intervals are smaller than the Simultaneous T^2 C.I.s.

Individual	X_1 (Sweat rate)	X_2 (Sodium)	X_3 (Potassium)
1	3.7	48.5	9.3
2	5.7	65.1	8.0
3	3.8	47.2	10.9
4	3.2	53.2	12.0
5	3.1	55.5	9.7
6	4.6	36.1	7.9
7	2.4	24.8	14.0
8	7.2	33.1	7.6
9	6.7	47.4	8.5
10	5.4	54.1	11.3
11	3.9	36.9	12.7
12	4.5	58.8	12.3
13	3.5	27.8	9.8
14	4.5	40.2	8.4
15	1.5	13.5	10.1
16	8.5	56.4	7.1
17	4.5	71.6	8.2
18	6.5	52.8	10.9
19	4.1	44.1	11.2
20	5.5	40.9	9.4